

Syed Muhammad Ayyan Ibrar

AI/ML Engineer | Python, Data Pipelines & Backend Systems

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WORK EXPERIENCE

Freelance AI/ML Engineer

Self-Employed

Remote

2024 – Present

- Built end-to-end Python data and ML pipelines covering data ingestion, preprocessing, validation, model training, evaluation, and production deployment.
- Designed data-processing workflows to transform and prepare heterogeneous datasets for downstream machine learning and AI systems.
- Developed backend services and automation workflows integrating data processing, databases, and model inference into production systems.

PROJECTS

Contract Agentic Society (CAS) — Lead Flagship

Autonomous Multi-Agent Federation for B2B Contract Lifecycle & Governance

Tech Stack: Python 3.12, Google Gemini API, FastN MCP, FastAPI, PostgreSQL, Asyncpg, Docker

- Isolated contract extraction from raw prompt text (`_extract_contract_text`) to bring End-to-End Benchmark Accuracy from 70.0% (35/50) to 100.0% (50/50).
- Eliminated false-positive clause leaks and recalibrated risk thresholds to bring Failed Cases from 15 to 0 and Risk Precision from 67.5% to 100.0%.
- Aligned event-type escalation triggers (`DISPUTE_RAISED`, `CROSS_SOCIETY_CONFLICT`, `TERMINATION_RISK_DETECTED`) to bring HITL Escalation Accuracy from 76.0% to 100.0%.
- Implemented full corporate governance policies (`RULE-001`–`RULE-006`) to bring Compliance Accuracy from 47.5% to 100.0%.
- Removed phantom clause-reference leaks (e.g. Section 8.2) from clean contracts to bring the False Positive Rate from 56.5% to 0.0%.

B2B Deal & Procurement Negotiation Engine

Autonomous Counterparty Negotiation & Concession Simulator

Tech Stack: LangGraph, Groq, FastAPI, Redis, Qdrant, PostgreSQL, LangSmith

- Designed an autonomous multi-agent bargaining engine that models vendor concession velocity and counterparty strategies to optimize commercial agreement terms.
- Formulated an asynchronous 17-message negotiation protocol with state verification, deadlock prevention, and cooperative coalition formation to resolve multi-stakeholder procurement impasses.
- Built dynamic trust modeling tracking counterparty behavioral signals across deal sessions, penalizing contradictory concessions and adversarial positioning.
- Benchmarked deal outcomes against traditional heuristic and centralized baselines, improving utility efficiency and fairness across simulated enterprise vendor disputes.

High-Stakes Risk Verifier & Epistemic Gatekeeper

Real-Time Hallucination Prevention & Safety Layer for Enterprise Agents

Tech Stack: LangGraph, FastAPI, Redis, Qdrant, PostgreSQL, LangSmith

- Developed a deterministic safety gatekeeper that monitors agent reasoning fidelity in real time to prevent high-liability automated actions in enterprise workflows.
- Built a 3-signal epistemic classifier that scores claim certainty and halts execution whenever confidence falls below calibrated thresholds, eliminating hallucinated output.
- Validated confidence calibration using Expected Calibration Error (ECE) and reliability curves, ensuring agent refusal behaviors strictly align with actual failure rates.
- Implemented a 3-tier memory system (Redis working memory, Qdrant vector retrieval, PostgreSQL audit logs) with full LangSmith tracing for enterprise auditability.

AuraSight

Edge-Native Multimodal Transaction & Voice Accounting System

Tech Stack: Android, Kotlin, Jetpack Compose, Gemma AI, On-Device Whisper (sherpa-onnx), SQLite

- Engineered a fully localized voice-first retail transaction assistant enabling visually impaired shopkeepers to independently manage billing, pricing, and credit ledgers (Khata).
- Implemented a constrained tool-calling architecture enforcing deterministic schema execution for financial calculations and inventory updates, eliminating arithmetic hallucinations.
- Optimized latency with on-device speech-to-text (Whisper ONNX) and physical button hardware triggers, reducing friction in low-connectivity retail environments.

EDUCATION

The Islamia University of Bahawalpur

BS, Artificial Intelligence (CGPA: 3.76)

Bahawalpur, Pakistan

2022 – 2026

- Advanced AI Bootcamp, GIK Institute (GIKI), Topi, Swabi — Grade A — LLM Engineering, RAG, Agentic AI, MLOps

SKILLS

Programming & Backend: Python, FastAPI, Pydantic, Git, Docker

Data Engineering: Data Ingestion, Data Cleaning, Data Validation, Data Transformation, ETL Pipelines, Unstructured Data Processing

Databases: SQL, PostgreSQL, ChromaDB, Qdrant, Redis

Cloud & AI/ML: Azure, GCP, Machine Learning, RAG, Embeddings, LLM Applications, Model Evaluation